

Logic Sim — User Manual (v1.19.0)

Logic Sim is an offline simulator and viewer of the plant DCS logic drawings: the digital interlock pages (DITL) and the 54 analog control sheets (ABC). It runs from one file, with no network, no OPC and no connection to the plant. It is for study, training and checking logic. It is **not** connected to the real DCS and never writes to it.

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1. Installation

Download from **GitHub → Releases** (take the latest `vX.Y.Z`). All three files have the same version in the name:

File	For	Install
<code>logic-sim-vX.Y.Z.html</code>	Any computer with Chrome / Edge	None. Double-click, or drag into the browser. Works offline.
<code>logic-sim-vX.Y.Z-portable.exe</code>	Windows	None. Portable: no installer, no administrator rights. Double-click to run.
<code>logic-sim-vX.Y.Z.apk</code>	Android	Copy to the phone/tablet, open it, allow "install unknown apps" for your file manager when asked.

Notes

- **Windows SmartScreen** may say "Windows protected your PC" because the exe is not code-signed. Click *More info* → *Run anyway*. The app has no network access (it blocks all of it).
- **Android** may warn about an unknown app for the same reason. The apk is signed with a sideload key, not a Play Store key.
- Updating = download the new version and use it. Your settings are not inside the program: save a **project file** (section 3.6) and open it in the new version.
- Only one version is needed. Old versions can be deleted.

2. Familiarization

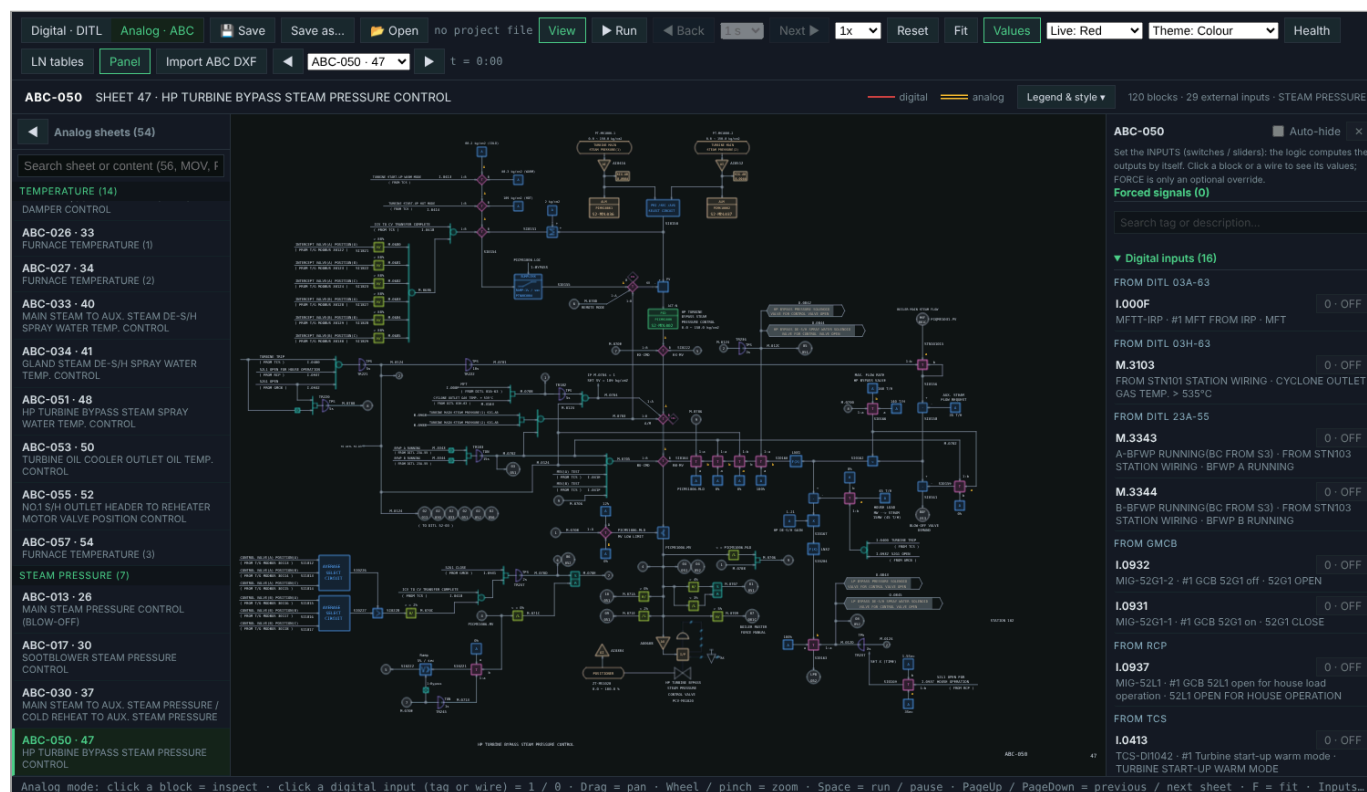
2.1 The two pages

The top-left switch chooses the page:

- **Digital · DITL** – the digital interlock logic pages (gates, flip-flops, timers). Read-only drawing behaviour; this page is never modified.
- **Analog · ABC** – the analog control sheets: controllers (PID), selectors (auto/manual and T switches), transmitters, valves/actuators, compensation and linear tables, with the digital permissives feeding them.

The app always **opens in VIEW mode** (section 2.5). Nothing is simulated until you press Run.

2.2 The analog screen



Top bar, left to right:

Control	Meaning
Save / Save as... Open	Save or open a project file (Ctrl+S). Keeps inputs, forces, switches, settings and your linear-table edits.
View	View mode (default): see 2.5.
Run / Pause	Start / pause the simulation (Space). Run from VIEW starts the simulation.
Back · 1 s...60 s · Next	While paused: step the time forward or roll it back by the selected amount.
1x...300x	Simulation speed. Each PID panel suggests a realistic speed for its loop.
Reset	Put everything back to the drawing's start state (inputs, forces, timers). Your linear-table edits are not cleared; they have their own reset.
Fit	Fit the sheet to the screen (key F).
Values	Show/hide the live values on the drawing.
Live / Theme	Colour of live signals; <i>Colour</i> theme or the muted <i>Engineering</i> theme.
Health	Reader health report of the sheet.
LN tables	The F(X) linear tables and their graph.
Panel	Show/hide the right panel.
Import ABC DXF	Load updated drawings; kept in this browser (see 3.9).
Imports	List of imported drawings, drawing-change report, Back to built-in.
DITL signals	The signals that come FROM / go TO the DITL page: one-click inputs and live values (3.4c).
HMI	The HMI graphics view: widgets connected to addresses; Tab / Float / Split (3.4d).
sheet	Previous / next sheet. PageUp / PageDown also work.

Left list: the 54 sheets grouped by function, with a search box (sheet name, or any text in the sheet).

Right panel: lists the sheet's **inputs** (digital switches, analog sliders / number boxes), forced signals, and – when you select something – the details of that block or wire.

2.3 Colour language

- Digital wire: thin line, live colour when **1**, grey when 0. Analog wire: thin solid line, always coloured (it carries a value, even 0); only in VIEW is it grey (the double-line tube is still available in *Legend & style*).
- Not-selected leg of a T (A/M) switch is grey; the selected leg is lit. Each leg is judged separately.
- A forced signal is dashed and carries a badge.
- Run and Pause use the same colours. View is all grey (see 2.5).
- Values are green by default (changeable in *Legend & style*).
- Valve/actuator: green = closed, red = open, blue = in between, white blinking = moving.

2.5 The three modes

Mode	What it does
VIEW (default when the app opens)	Every wire is grey so the lines are easy to read. Nothing is simulated and no input or force can be changed. Click a block or a wire to trace it automatically (see 3.4); click an empty place to clear the trace.
RUN	Press Run (or Space). The wires that should be live light up. Starting from VIEW changes nothing by itself: no value is forced. Inputs and forces exist only after you set them; Reset removes them all.
PAUSE	Same colours as RUN, but frozen. Choose a step (1, 5, 10, 30 or 60 s) and press Next ► to run the logic forward by that time, or ◀ Back to roll it back by that time.

Press **View** to return to VIEW at any time (it also stops the run; your values are kept and shown again when you go back to RUN).

2.4 Addresses and descriptions (unit 1)

Tags such as I.0413, M.3103, TR0708, S1 M.0160 are looked up in the unit-1 IO list and memory lists that are built into the app. When you select a tag (digital) the tag is highlighted **yellow**; for analog it is **sky blue**; and a small card shows *tag name [address · station]*, the description and the type. Hover the mouse on ANY address text of the drawing (SI, B, I, O, M, TR...) and the same card appears; select a block that carries an address (for example SIG.AB) and its card is shown. The input list in the right panel also shows the description under each tag, and the search box finds by description.

3. How to use

3.1 Open a sheet and run it

1. Choose **Analog · ABC**, pick a sheet from the left list (e.g. ABC-050).
2. Set the inputs in the right panel: click a digital input (or the tag on the drawing) to toggle 1/0; type a value or drag the slider for analog inputs.

3. Press **Run** (or Space). Watch the outputs. Press **Pause** to stop.
4. Origin signals (inputs) are set by you; driven signals are computed by the logic. **FORCE** (in the panel, for any selected wire) is an optional override of a computed value – use it for testing only; *Release* removes it.

3.2 Step through the logic (paused)

Press **Run**, then **Pause**. Choose 1 / 5 / 10 / 30 / 60 s and press **Next** ► or ◀ **Back**. The colours are the same as in RUN, so you see exactly what turned on or off. Back restores the whole simulation state of that moment (values, timers, valves). It keeps about the last 10 minutes of simulated time; a Back that is farther than the history goes to the oldest point. Inputs you changed after that moment are not replayed.

3.3 Select and understand a block

Click a block or a wire. The right panel shows its inputs and outputs with live values, its parameters (timer seconds, PID tuning, ranges, ...) and a **Why?** line that explains the present output in words (e.g. "AND = 0: M.3103=0 is 0 (one 0 is enough)"). For some kinds the explanation is general ("Inputs now ... → output ...") because a detailed text is not written yet.

3.4 Trace (VIEW mode)

1. In VIEW, click a wire or a block. The trace appears by itself: a **glow** is drawn around the wires, white = what you selected, **blue = driven by** (upstream), **magenta = feeds** (downstream); the rest is dimmed.
2. The panel lists each wire with its name, description and value. Click a row to move the selection there.
3. For T/A-M switches only the leg in use is followed. *Selected T leg only / Both T legs* follows both.
4. **Leaving the sheet.** Under the title *Trace* the panel lists **every exit** of the selected signal ("This signal leaves the sheet (n): ► ABC-004A (circle 12) → receiving wire · description · feeds k"), not cut by the 60-row limit of the lists. Each wire in the lists that continues in another sheet also has a pink row ► **continues in sheet ABC-xxx (circle n) → wire · description · feeds k blocks**. One click opens that sheet with the receiving wire(s) selected and the trace goes on there. A circle that says "(TO ABC-001D) (TO ABC-020)" gives one row per sheet. A grey row says → **to DITL pp-nn** (the signal leaves to the DITL page: not simulated) or ◀ **from DITL pp-nn**. A row ◀ **sheet ABC-xxx** means the input comes from another sheet: click it to jump to the sending wire.
5. **Works in RUN and Pause too.** In RUN / Pause (no Trace lists) the selection panel has a block **Signal path** with the same **Path** crumbs, the list of **every exit** of the selected signal, and the **Map** button; the simulation keeps running while you move from sheet to sheet (all linked sheets run together). **Path (breadcrumbs)**. After a jump the panel shows **Path: ABC-003B › ABC-003A › ABC-002 › ABC-004A**. Click any sheet name to return there in ONE click (the path after it is dropped), ↻ **Back to start** returns to the first sheet and its wire, **Clear path** forgets it. A sheet that is already in the path is not added again: when the signals go round in a circle (A → B → A) the path is cut back, it never grows without end. ↺ Back still goes one step at a time.

6. **Map: where does this signal go (all sheets).** One button shows the whole cross-sheet downstream of the selection in ONE list: indented by hop, one line per sheet (a sheet reached again is shown once as ↻), each line = sheet · receiving wire · description · what it feeds (block kinds). Click a line to go there; the path is kept.
7. Click an empty place of the drawing: the trace goes away and you are back in plain VIEW.

3.4b Trend (live graph of a PID, MAN, SUMA / SUMP or FX)

Select a **PID, PIDV, MAN, SUMA / SUMP (integrator) or FX (linearizer)** block (in VIEW, RUN or Pause): the panel shows a **Trend**: a small live chart with the name and live value of each trace. PID / PIDV: **SV, PV, MV** and the P, I, D parts and the error; the PV is the input of the DEV block that the drawing marks "PV" (not always the minus pin). **No Trend for a wire, a digital signal or any other block** (user rule: there is no history, so a graph of a digital signal is useless). Zoom opens a floating window with a control strip.

3.4c DITL signals (button)

The DITL page is not simulated with the ABC sheets and is not touched. The signals that the ABC sheets say come **FROM DITL pp-nn** are inputs of the ABC set: the button **DITL signals** lists them (sheet, DITL reference, wire name and description, live value) with a **one-click control** (switch 1 / 0, or a number for an analog input). It works in RUN and Pause, all linked sheets run together, and everything that the ABC logic computes stays under the control of the logic (it is not in the list). The signals that go **TO DITL pp-nn** are listed with their live value (read-only). Texts that cannot be tied to a wire are counted and named at the top of the list.

3.4e Faceplate values, COS manual control, plant model, Trend (v1.17.0, updated after v1.18.0)

- **Faceplate values:** the PID / MAN / SUMA values written from the DCS ($K_p = 100$ / P, T_i , T_d , limits, ranges, unit) are applied automatically; the panel says "FILE" for them and "ASSUMED" for what is still a guess (docs/ASSUMED-VALUES.md).
- **COS** (operator manual value / change-over): in the panel section "COS manual values and change-overs": slider + number, range and unit from the diagram, "target → now"; greyed out when the COS is not energized by the switching logic (exception: a COS used as SV or PV).
- **Plant model** (simulation only, in the controller panel): the plant answers where the field answers: it writes the TRANSMITTER (AI) of the PV, or the input wire of the PV; the blocks between the transmitter and the PID run as drawn. The PV and the transmitter are read only (greyed, with the reason); control the SV. Gain / time constant / dead time are ESTIMATED from the tuning of the controller (docs/PLANT-MODEL.md lists every controller: transmitter / input / pin / shared / none). Two controllers on the same measurement share one model. Untick the model to use the PV as an input again.
- **FORCE / SIM wins over the plant:** a point that you hold (F in the HMI, or FORCE in the panel) is never changed by the plant, the SV or the manual command; the plant keeps running underneath and the value goes back to it when you release.
- **Trend** only for PID, PIDV, MAN, SUMA and FX: Process pane (SV, PV; the PV is the pin that the drawing marks "PV", not always the minus pin) and Controller pane (MV, P, I, D, error); click a name to hide it; Zoom opens a floating window (drag the bar, corner resize,

transparency, wheel zoom) with a control strip to change Kp, Ti, Td, limits, SV and the plant model without leaving the graph. No Trend for wires, digital signals or other blocks.

- **Panel:** fixed width (380 px), every text wraps. The arrows of a number box apply the value like Enter.
- DITL signals: the list shows the open sheet; "All sheets" shows everything.

3.4f Who controls: the HMI, FORCE / SIM (WIP after v1.18.0)

While the **HMI is open it is the controller**: the panel and the diagram cannot write inputs or forces (the controls are greyed and a banner says so); close the HMI to use them again. In the HMI every value has an **F** box: press it to hold the value (SIM on the field side, FRC on the logic side, blue P = driven by the plant); then the slider / button sets the held value; F again releases. It works in RUN and in PAUSE, for analog and digital. The Auto page lists every transmitter, signal-bad flag and input of the open sheet, the plant values (read only, F to hold) and the outputs to the DITL.

3.4d HMI view (graphics page connected to the addresses)

HMI (toolbar) opens a graphics page whose buttons, lamps, numbers, sliders and faceplates are **connected to addresses**. It runs on the SAME engine as the diagrams: press an HMI button and the diagram reacts; the HMI shows the same values. Three display modes (buttons at the top right of the HMI):

- **Tab** – own view: the HMI replaces the diagram area (like Digital · DITL / Analog · ABC); the right panel stays.
- **Float** – a window over the diagram that you can **move** (drag its title bar), **resize** (drag the corner); the diagram keeps running behind it, so you can press an HMI button and watch the diagram.
- **Split** – the HMI beside the diagram with a **draggable divider**; nothing is covered.

Widgets (Edit → "+ Add widget...", then click on the page): Text, Lamp, Number, Bar, Button (toggle, or *momentary* = 1 while pressed), Slider (analog input), Valve, Motor / fan, Tank (level), Pipe, Trend (up to 3 addresses), Faceplate (tag of a PID / PIDV / MAN: SV, PV, MV bars and numbers; a MAN has a slider for its value; a PID with an input SV has an SV slider; "open ►" jumps to the block in the diagram).

Address = a tag or address written on the drawings (M.0097, SI0150, AI0130, B.0646, FICFA1043B.MV ...) or a **DITL reference** ("DITL 13-69"). The status line under the page says where it was found (sheet, wire, INPUT or computed). If the address exists on several sheets, choose the sheet in the *Sheet* box (auto = the best place: for a button the input of the ABC set, for a lamp the wire that is computed).

Buttons and sliders: on an **input** (signals FROM DITL and other external inputs) they write the one-click input. On a wire that the logic **computes** they **FORCE** it: the button shows "force" / "FORCED" (shift + click releases the force). Momentary buttons release on mouse-up.

Edit mode: click a widget to select it, drag to move (grid of 5), change label / address / min / max / colour / size in the line under the page; **Duplicate**, **Delete** (or the Delete key). **+ Page**, **Rename**, **Delete page**. **Background...** puts a picture (for example a screenshot of the real HMI) under the widgets; the page takes the size of the picture (largest side 1600). **Auto: this sheet** makes a new

page from the sheet you are viewing: a faceplate for every PID / MAN and a button (digital) or slider (analog) for every signal that comes FROM DITL.

Saved with the other settings (browser storage and the project file), restored after F5. Every sheet that a widget reads or writes keeps running while the HMI is open (all linked sheets run together). The DITL page is not touched.

HMI v2 (v1.18.0): Auto page (Edit → Auto page) now builds a page of the open sheet with every faceplate and EVERY manual input (analog slider with the range of the drawing, digital button), plus the inputs that come from other sheets (read only, with the sheet name) and the PV that the process model simulates (read only: set the SV). **Find...** searches every tag / address with its description; **Pick on diagram** takes the address of the wire you click. **Zoom** with the wheel or + / – / Fit, **pan** by dragging the empty page, widgets move smoothly (tick Snap for a 5 px grid), **Lock** stops editing but you can still operate. A / M / CAS are not buttons of the HMI: they come from the switching logic, whose inputs (tag.MAN / .LOC / .REM) are among the digital buttons.

3.5 PID controllers

Select a PID block. The panel shows PV, SV, output, A/M mode and tuning (Kp, Ti, Td), direct/reverse action, and the range read from the drawing. Default tuning is chosen by the loop type from the tag (flow, pressure, temperature, level, analysis, speed) and a **Use this speed** button sets the simulation speed that suits the loop. These are typical values, **not** the plant's real tuning: replace them when you have the real data.

3.6 Project file (save and restore your work)

- Press  **Save** (Ctrl+S). The first time it asks where; after that it overwrites the same file. *Save as...* makes a new file;  *Open* loads one.
- The file stores inputs, forces, switches, sliders, settings and linear-table edits for all sheets, and the current sheet. It also remembers which version saved it.
- The app also keeps a copy in the browser's storage, but a project file is the only copy you control: save one before changing computer, browser or version.
- Windows exe: Save/Open use normal file dialogs. Android: Save shows the share sheet (save to Files/Drive/...), Open picks a file.

3.7 LN tables (F(X) linearization)

The 89 DCS linear tables of LINEAR.xls are built in, plus S1-LN38 and S1-LN39 of the drum level (from the Drum Level Calculation file). **LN tables** shows the graph. **Edit table** lets you change the LX/LY points (in percent of the ranges; X and Y are computed). A warning shows if LX goes backward. Edits have their own **Reset this table** and **Reset ALL linear edits**; Reset on the sheet does not clear them. Each FX block finds its table by station and LN number; if only another station has that number, a warning is shown in the FX panel instead of silently using it.

3.8 View mode

See 2.5 and 3.4. In VIEW nothing can be changed by accident: you can select, search, trace and read the descriptions.

3.9 Updating drawings (Import ABC DXF) — v1.16.0

Import ABC DXF loads updated analog drawings (the sheet name is the file name, e.g. `ABC-050.dxf` replaces ABC-050). The imported file is **kept in this browser** and used again every time you open the app. The saved values (forces, inputs) of that sheet are cleared at import, because the numbers of the wires can change. After the import the **Imports** panel opens with the **drawing-change report**: counts of blocks / nets / lines / circles / texts (built-in vs imported), the block kinds that changed, and the blocks and texts that exist in only one of the two. "IDENTICAL" means nothing changed. **Back to built-in** removes the import and returns to the built-in drawing. Check **Health** afterwards and keep a saved project file before importing.

3.10 Assumed values and data files

Some numbers are **not written on the drawings**. The button **Assumed values** (Analog bar, next to *LN tables*) lists all of them, and the panel of the block says it too:

- **TP** (flow temperature compensation, 8 blocks): the operating temperature comes from your Compensation file (302 / 35 / 91 / 287 °C). Real data. $DP' = DP / Kt$.
- **ALM** (104 alarms): the four limits HH / H / L / LL are **ASSUMED** for a 150 MW CFB boiler with reheater. They are NOT the DCS values. Type your own value in the block panel to replace one (it is saved with the project).
- **Ramp boxes** (8 boxes without a rate on the drawing) and **PO / PIDV pulses** (cycle 2 s, full stroke 60 s, shortest pulse 0.2 s): **ASSUMED**, editable in the panel.
- **F(X) LN38 / LN39 of ABC-010** (drum level compensation): taken from your Drum Level Calculation file because LINEAR.xls has them only for station 2.

Every number, with the reason for it, is in `docs/ASSUMED-VALUES.md`; the files you sent and where each one is used are in `docs/DATA-FILES.md`.

3.11 Keyboard

Space run/pause · PageUp/PageDown or $\leftarrow/\rightarrow$ sheet · F fit · Ctrl+S save · wheel/pinch zoom · drag pan.

4. Troubleshooting

Problem	What to do
The page is blank or the sheet list is empty	Wait a few seconds on first open (the data unpacks). Use a current Chrome/Edge. Do not open the html from inside a zip: extract first.
Windows says "protected your PC"	Normal for an unsigned exe. <i>More info</i> → <i>Run anyway</i> .
Android will not install	Allow "install unknown apps" for the app you opened the apk from. If an older Logic Sim is installed from a different key, uninstall it first.
Nothing moves / all wires are grey	The app opens in VIEW: press Run. Valves, timers and ramps move only while running or while you press Next ►.
I changed an input but the output did not change	Output may be waiting on a timer, or is forced (dashed wire / <i>Forced signals</i> count). Release the force.
A typed value jumps back	Press Enter or click ✓ after typing; the box keeps your number until then.

My settings are gone	The browser storage was cleared or blocked (private window). Open the saved project file. The status text in <i>Legend & style</i> → <i>Saving</i> tells which storage works.
Reset removed my forces but not my linear edit	By design: linear edits have their own reset.
Values I saved in an older version are gone	When a new version changes how the drawings are read, the saved values of the affected sheets are dropped (the settings stay). Re-enter them and save a new project file.
A warning appears in an FX block about a station	The LN table number exists only in another station; the app used it and warns you. Check the drawing.
Description missing for an address	The IO/memory lists cover unit 1 only. Some labels are not in the lists.
Something else	Note the sheet name, what you clicked, and a screenshot, and report it.

5. What is and is not simulated

Checked against the drawings and your files (v1.15.0): flip-flops (reset wins when $S = R = 1$) and timers follow the symbol list of ABC-000; 150 of 153 timers equal the TR table of your memory lists (type and time); 157 of 157 AI ranges equal your IO list; in the simulation every digital output of the 51 sheets can be driven to both 0 and 1 (2191 of 2192; 18 need a sequence of input changes, e.g. latches and pulses); every link between sheets carries its value; no wire is a dead end except the real exits (TO DITL / TCS, annunciator, memory bit); on the screen a digital wire is lit only when its value is 1. The ABC → DITL crossings are not simulated (the DITL page is never touched). This shows the logic is read completely, not that it matches the plant: sheet-by-sheet comparison with the PDFs is still open.

Simulated: gates, flip-flops, timers (TON/TOF/pulse), T/A-M selectors, comparators, SEL (average of healthy transmitters), CTK, PID with typical tuning, valves/actuators with travel time, ramps, linear and compensation tables, constants, alarms (limits assumed), pulse output.

Not (yet): the real plant response (there is no plant model: you set the PV yourself or through a simple model), real PID tuning and alarm limits of the plant, other units than unit 1 for descriptions, communication with the real DCS. A few drawing symbols are still unrecognised (listed in docs/BACKLOG.md). Treat results as a study aid and verify against the real system before acting on them.